Chapter 05: Results And Evaluation

This chapter presents the outcomes of the Smart Waste Detection and Segregation Platform, evaluates the performance of the MobileNetV2 model, and compares the system with existing waste classification approaches.

5.1 Results

The Smart Waste Detection and Segregation Platform successfully detects waste categories from static images and real-time camera input. The system evaluates images across 10 waste classes:

- battery
- biological
- cardboard
- clothes
- glass
- metal
- paper
- plastic
- shoes
- trash

The model performed well in classification after being trained using MobileNetV2 on augmented data.

5.1.1 Model Training Results

Overall Accuracy: 94%

The accuracy greatly increased after the model was adjusted because of:

- data augmentation
- unfreezing top MobileNetV2 layers
- using a smaller learning rate (1e-5)

Training Accuracy Curve:

- Training accuracy exhibited a steady rising trend.
- Without significant overfitting, validation accuracy stabilized.

Loss Curve:

- Over the course of epochs, loss slowly diminished.
- Robust generalization is indicated by validation loss remaining near training loss.

5.1.2 Prediction Results

The average prediction time for each image is between 0.9 and 1.2 seconds.

(using CPU, no GPU acceleration)

Confidence Score Range:

Strong model certainty was demonstrated by the majority of predictions, which yielded confidence scores above 0.85..

Example Predictions

Table 5.1 - Prediction result table

Input Image	**Output Class**	**Confidence**
Plastic bottle	plastic	0.97
Old t-shirt	clothes	0.92
Cardboard box	cardboard	0.95
Broken wine glass	glass	0.91

5.1.3 Real-Time Camera Detection

The real-time camera classification was tested under different conditions:

Table 5.2- Camera Performance under various condition

Condition	**Result**
Bright lighting	Smooth detection
Indoor lighting	Slight drop in confidence but accurate
Moving objects	Frame-by-frame detection successful

Low-light	Model still predicts but confidence drops

Average FPS: 5–10 FPS (depending on lighting and CPU power)

5.1.4 System Output Interface

For every prediction, the UI displays:

1. Waste class
2. Confidence percentage
3. Carbon footprint info
4. Proper disposal method
5. Reuse / recycling suggestions

This transforms the model prediction into environmentally meaningful insights, making the platform more educational and practical.

5.2 Interpretation of Results

1. MobileNetV2 performs exceptionally well on waste classification

- Lightweight
- Fast inference
- High accuracy
- Works well even with limited dataset sizes

2. Data augmentation improved generalization

By applying:

- rotation
- zoom
- shear
- horizontal flip
- brightness shifts

The model handled real-world variations well and prevented overfitting.

3. utility is also greatly improved by adding the feature of camera detection

Without having to manually upload images and then check , user may quickly identify waste.

4. The backend (FastAPI) performs quickly and steadily.

- Low latency
- No memory overload
- API responds correctly to incorrect inputs

5. 5. The UI is responsive and quite easy to use

The frontend in React offers:

- smooth preview image
- categorizes in real time
- Information on environmentally friendly waste

5.3 Comparison of Current Solutions

Numerous current studies and public implementations make use of:

1. TrashNet Dataset (Baseline Model)

Accuracy typically ranges from 75%–87% using CNN, VGG16, or ResNet.

Our system outperforms them with:

- 94% accuracy using MobileNetV2
- Fine tuning on augmented data

2. Heavier Models Used in Literature

Examples:

- ResNet50
- VGG19
- EfficientNet

These models:

- require GPU
- slower on CPU
- not suitable for real time classification

MobileNetV2,selected for this project:

- Lightweight
- Faster inference

- Runs on mobile and edge devices
- Suitable for real-time camera use

3. Government and Industry Waste Apps

Apps like KachraSetu, Swachhata app, Bin-e use detection but:

Table 5.3 - Feature comparison with existing apps

Feature	**Existing Apps**	**Our System**
Real-time camera waste detection	Partial/limited	Fully implemented
AI model transparency	Not provided	Open architecture
Confidence score	No	Yes
Disposal & recycling tips	Basic	Detailed environmental impact insights
Open-source orientation	Mostly closed	Fully open for upgrades

4. Web-Based Classification Tools

Most existing tools:

- Only allows image uploading and generally has no real-time camera
- They also provide only predictions and generally don't give insights about the environment.

Our platform adds and provides:

1. Real-time classification
2. Waste management guidance
3. Carbon footprint estimation
4. Smooth React–FastAPI integration

Summary of Comparison

Table 5.4 - comparisons

Aspect	**Existing Systems**	**Our System**
Model	Heavy CNNs	Lightweight MobileNetV2
Accuracy	75–88%	94%
Speed	Moderate	Fast (<1.2s)
Camera support	Limited	Full real-time camera
Disposal info	Rare	Included

Confidence score	No	Yes
Educational purpose	Minimal	Strong